Objective Entropy in Ontology

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Abstract

This paper introduces the formulation of multiplicative entropy proposed by Zou (2025) from the perspective of objective ontological entropy[1].

Traditional Boltzmann-Gibbs statistical entropy bears an epistemological background: entropy depends on artificial coarse-graining and the selection of macrostates, and entropy increase is merely a statistically probabilistic conclusion rather than an inevitable outcome of underlying dynamics. Based on the discrete Space-Elementary-Quantum (SEQ) network model, this paper introduces objective ontological entropy, defined as the product of node energies. It constructs an underlying thermodynamic picture free from observer-dependence and coarse-graining assumptions, featuring strictly monotonic entropy increase constrained by energy conservation. In this model, the Planck time serves as the minimal update cycle, and energy transfer constitutes fundamental events. Time itself is defined as the count of irreversible energy-redistribution events within the network. By comparing entropy-production growth rates (instead of absolute entropy values), the logarithmic entropy-production rate of this ontological entropy approximates the Clausius entropy-production rate of heat conduction under small-gradient energy-(temperature) distributions. It offers a new alternative perspective bridging underlying discrete dynamics and macroscopic thermodynamics.

Keywords: ontological entropy; objective entropy; SEQ; Space Elementary Quantum; entropy-production rate; non-statistical thermodynamics; ontological thermodynamics; irreversible processes

1. Introduction

The second law of thermodynamics is the core law describing irreversible processes. Nevertheless, mainstream statistical mechanics has long confronted fundamental epistemological dilemmas. For Boltzmann entropy $S = k^B \cdot \ln\Omega$, the core quantity Ω counts microstates and relies on artificial coarse-graining of phase space. The Gibbs mixing paradox further demonstrates that entropy values vary with the observer's power of discrimination.

Since microscopic dynamics is time-reversal-invariant, the Loschmidt reversibility paradox and the Poincaré recurrence paradox indicate that entropy increase within the Boltzmann framework is not strictly necessary. It represents only a highly probable statistical tendency for macroscopic systems; theoretically, spontaneous entropy-decreasing fluctuations are permitted.

Within the traditional statistical-thermodynamic paradigm, entropy tends to describe human ignorance about a system (an epistemological quantity), rather than an intrinsic ontological property independent of observation. Theoretical researchers have long sought a purely ontological definition of entropy: entropy shall be uniquely determined solely by the system's own physical state, free from observer choices, and entropy increase shall be enforced by underlying dynamics instead of statistical approximations. The Space-Elementary-Quantum (SEQ) discrete-network model[1] provides such a candidate approach.

2. The SEQ Model and Definition of Ontological Entropy

The SEQ model postulates that space consists of discrete elementary quantum nodes, each carrying integer-valued energy units.

Definition of Objective Ontological Entropy: $S = \prod_i m_i$

where m_i denotes the number of energy units carried by the i-th SEQ node.

For a closed network, total energy is strictly conserved: $E = \sum_i m_i = \text{constant}$

E stands for total conserved energy, and the total number of nodes N remains fixed.

From the Arithmetic-Mean-Geometric-Mean (AM-GM) inequality under fixed total energy: $S = \prod_i m_i \leq (E/N)^N$

Equality holds when all nodes possess identical energy, corresponding to thermodynamic equilibrium.

Distinct from conventional thermodynamics: the upper bound of entropy is directly derived from energy conservation, without invoking the equal-a-priori-probability postulate as an extra statistical assumption.

2.1 Underlying Dynamical Rules

Energy transfers occur only between adjacent SEQ nodes subject to the transfer condition: $m_i > m_j + 1$

When satisfied, one unit of energy flows from the higher-energy node i to the lower-energy node j: $m_i \rightarrow m_i - 1, \quad m_j \rightarrow m_j + 1$

The entropy ratio after a single valid transfer reads: $S'/S = [(m_i-1)(m_j+1)]/(m_i \cdot m_j) = 1 + (m_i-m_j-1)/(m_i \cdot m_j) > 1$

Every valid energy transfer strictly increases ontological entropy; spontaneous entropy reduction cannot occur.

2.2 Time: Event Counting Rather Than Background Container

Postulate: each valid energy transfer triggers one state update of the SEQ network. A single update occupies one Planck-time unit t_p.

Time is not a pre-existing continuous background coordinate. **Physical time is the count of irreversible energy-transfer events inside the SEQ network**.

Independent node-pairs can perform energy transfers in parallel within one Planck-time cycle. In local regions with no valid energy-transfer events, local event counting halts. This yields a microscopic mechanism for gravitational time dilation: reduced update frequency of local SEQ network slows local physical clocks.

3. Entropy-Production Growth Rate: Connecting to Macroscopic Clausius Thermodynamics

This framework does not require the absolute values of ontological entropy to match Clausius thermodynamic entropy. Instead, the **relative growth rate of entropy (entropy-production rate)** — the dynamical rate of irreversible evolution — is selected as the core benchmark for comparison with macroscopic experiments.

Ontological entropy takes a product form. Logarithmic transformation converts it into a summable quantity and circumvents explosive growth of raw entropy values. Every valid energy transfer yields a definite logarithmic entropy increment. More energy-transfer events per unit time correspond to higher entropy-growth rates.

The entropy-production behaviour yielded by this ontological-entropy model approximates the structural behaviour of Clausius entropy production in classical heat conduction. The larger the gradient, the greater the entropy production..

Note: This alignment with macroscopic Clausius entropy-production holds under extrapolated small-gradient conditions. When energy-distribution gradients become large, the matching condition breaks down.

4. Core Comparison: Ontological Entropy versus Boltzmann Statistical Entropy

Dimension	Boltzmann Statistical Entropy	SEQ Objective Ontological Entropy
Nature of entropy	Epistemological quantity; depends on coarse-graining and observer's	Ontological quantity; uniquely determined solely by nodal

	partitioning	energy distribution of the system
Behaviour of entropy increase	Statistically probable increase; fluctuation-driven entropy decrease is permitted, not strictly monotonic	Strictly monotonic increase upon valid transfers; no spontaneous entropy decrease
Relation to energy conservation	Energy conservation acts merely as external boundary condition and cannot derive entropy upper bound	Energy conservation directly imposes theoretical entropy upper bound via the AM-GM inequality
Treatment of time	Continuous background parameter, independent of thermal-motion events	Time = count of irreversible energy-transfer events; no idle background time passage
Fundamental assumptions	Equal a-priori probability, molecular chaos, coarse-graining assumptions	Fixed node count, energy conservation, unidirectional gradient-driven transfer rules

5. Theoretical Potential, Open Questions and Verification Routes

5.1 Theoretical Potential

1. It eliminates observer-dependence of entropy at the fundamental level and realises ontological entropy. Irreversibility is embedded within underlying dynamics instead of emerging as a collective statistical effect of massive particles.

2. Under near-equilibrium small-gradient conditions, entropy-production rates show structural proximity to classical heat-conduction results. This opens potential pathways to bridge microscopic discrete networks and macroscopic thermodynamics.

3. The arrow of time, irreversible evolution and gravitational time-dilation are unified within a single network picture, offering expansion potential for fundamental-theory unification.

5.2 Outstanding Open Problems

Establish mathematically and phenomenologically consistent alignment with macroscopic Clausius temperature-gradient thermodynamics.

6. Conclusion

The ontological entropy defined within the SEQ discrete-network framework provides a novel alternative scheme distinct from epistemological statistical entropy. As an intrinsic system property free from observer-imposed coarse-graining, its entropy upper bound originates from

energy conservation, and valid energy transfers guarantee strictly monotonic entropy growth. Taking entropy-production growth rate as the comparison benchmark, it approximates Clausius heat-conduction entropy-production under near-equilibrium small-gradient conditions.

This theory remains at an early preliminary stage. Further work is required on mapping relations, dynamical equations, macroscopic-limit derivations and experimental comparisons. Nevertheless, it opens an ontological path to root thermodynamic irreversibility within underlying discrete space-elementary-quantum networks, offering promising new perspectives for re-examining entropy, the arrow of time and foundations of thermodynamics.

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